

Mark schemes

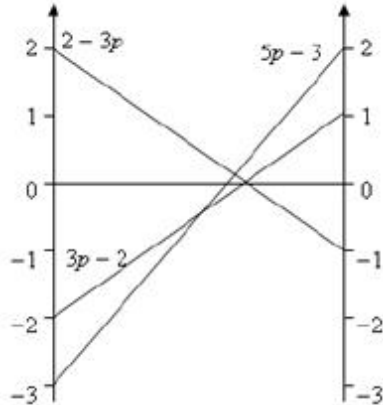
Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies correctly all three row minima or all three (two) column maxima	AO1.1a	M1	Row minima: -1, -2, (-1) Column maxima: 4, 0, (1)
	Determines correctly max(row min) and min(col max)	AO1.1b	A1	max (row min) = -1 min (col max) = 0
	Makes correct statement for the play-safe strategies of each player	AO3.2a	A1 CAO	Victoria plays <i>R</i> (or <i>P</i>); Albert plays <i>Y</i>
(b)	Identifies strategy and provides justification with reference to dominance	AO2.4	E1	Albert should never play <i>Z</i> because strategy <i>Y</i> dominates strategy <i>Z</i>

(c) (i)	States correctly that strategy <i>R</i> dominates strategy <i>P</i> (PI)	AO1.1a	B1	Victoria should never play <i>P</i> because strategy <i>R</i> dominates strategy <i>P</i>
	Introduces and defines a probability variable	AO3.3	M1	Let Victoria play strategy <i>Q</i> with probability p and strategy <i>R</i> with probability $1 - p$.
	Finds correctly both expected gain expressions for Victoria	AO1.1b	A1	If Albert plays: X: expected gain for Victoria $= -2p + 4(1 - p) = 4 - 6p$
	Clearly states that, as the game has no stable solution, the optimal solution occurs when the two expected gain expressions are set equal to each other, and then solves correctly 'their' equation constructed using 'their' expected gains for Victoria	AO2.2a	A1F	Y: expected gain for Victoria $= -(1 - p) = p - 1$ No stable solution, so optimal value of p occurs when: $4 - 6p = p - 1,$ $p = 5/7$
	Interprets correctly the solution to the problem in the context, giving the optimal mixed strategy for	AO3.4	E1	Victoria should play strategy <i>Q</i> with a probability of 5/7 and <i>R</i> with a probability of 2/7

	Victoria			
(ii)	Substitutes 'their' value of p into one of 'their' expected gain expressions, resulting in a value of the game	AO1.1b	B1F	$4 - 6 \times (5/7) = -2/7$ or $(5/7) - 1 = -2/7$
(iii)	Recognises correctly that Victoria can improve on the value of the game if Albert does not play an optimal mixed strategy	AO3.5b	E1	The expected pay-off for Victoria will only be $-2/7$ if Albert plays an optimal mixed strategy between X and Y
Total 11 marks				

Q2.

Marking Instructions	AO	Marks	Typical Solution
Deduces strategy C is a dominated strategy	AO2.2a	B1	$-3 = -3$, $-4 < -2$, $1 < 2$, hence strategy B dominates strategy C
Introduces and defines a probability variable	AO3.3	M1	Let John choose strategy A with probability p and strategy B with probability $1 - p$. If Danielle plays:
Finds correctly all three expected gain expressions for John	AO1.1b	A1	X: expected gain for John = $2p - 3(1 - p) = 5p - 3$
Constructs a graph with at least one vertical axis and plots one of 'their' expected gains correctly (PI)	AO1.1a	M1	Y: expected gain for John = $p - 2(1 - p) = 3p - 2$ Z: expected gain for John = $-p + 2(1 - p) = 2 - 3p$
Identifies correctly the optimal point of intersection from 'their' graph and finds 'their' value of probability variable	AO1.1b	A1	
Interprets correctly the solution to the problem in the context, giving the optimal mixed strategy for John	AO3.2a	E1	$3p - 2 = 2 - 3p$

			$6p = 4$ $p = 2/3$ John should play A with probability $2/3$ and B with probability $1/3$ (and C with probability 0)
Total 6 marks			

Q3.

(a) $R_C > R_B$

oe

E1

1

(b)

$$\begin{array}{c}
 A \\
 C
 \end{array}
 \left(
 \begin{array}{ccc}
 -2 & 0 & 3 \\
 4 & 1 & -1
 \end{array}
 \right)$$

K plays A prob p

C prob $1 - p$

P plays

D, K wins $-2p + 4(1 - p)$ ($= 4 - 6p$)

E, K wins $1 - p$

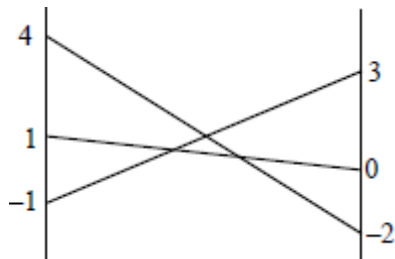
F, K wins $3p - 1(1 - p)$ ($= -1 + 4p$)

Allow 2 expressions in unsimplified form

M1

All 3 correct

A1



Must have 3 lines

M1

With values shown

A1

Max at $1 - p = -1 + 4p$

Identifying correct maximum from their graph

M1

$$p = \frac{2}{5}$$

(K plays) A prob $\frac{2}{5}$, C prob $\frac{3}{5}$

Both stated, coming from equating correct two equations and M2 scored

A1

Value of game = $\frac{3}{5}$

B1

7

(c) P plays D prob p

E " q

F " $1 - p - q$

K plays A, P loses

$$-2p + 3(1 - p - q) = 3 - 5p - 3q$$

Either (unsimplified) expression correct

M1

K plays C, P loses

$$4p + q - 1(1 - p - q) = -1 + 5p + 2q$$

$$3 - 5p - 3q = \frac{3}{5}$$

$$-1 + 5p + 2q = \frac{3}{5}$$

$$2 \quad -q = \frac{6}{5}$$

Equating BOTH of their expressions to value of their game

m1

$$q = \frac{4}{5}$$

Or for finding p

A1

CSO

$$5p + \frac{8}{5} - 1 = \frac{3}{5}$$

$$p = 0$$

P plays D prob 0

E , prob $\frac{4}{5}$

F , prob $\frac{1}{5}$

All three needed, must have scored previous A mark

E1

4

Alternative method

Probability of D is 0

OE, might be earned in final line

E1

$$3(1 - p) = \frac{3}{5} \quad \text{or} \quad p - 1(1 - p) = \frac{3}{5}$$

Or equating the expressions

(M1)

$$p = \frac{4}{5}$$

m1

E prob $\frac{4}{5}$ F prob $\frac{1}{5}$

(A1)

CSO

[12]

Q4.

- (a) R min $-4, -5, -2$ plays C
Either C or E stated

B1

- J max $4, 1, 3$ plays E
Both C and E stated

B1

and all values shown

E1

3

- (b) maximin R = $-2 \neq 1$ = minimax J
Correct values must be stated

E1

1

- (c) (For Juliet,) col E dominates col D

E1

1

- (d) (i) Signs changed as J gains = R losses

E1

Gains written as rows

E1

2

- (ii) Let J play E prob p

F $(1 - p)$

If R plays A, J wins $4p$

B $5p - 3(1 - p)$

2 correct expressions seen

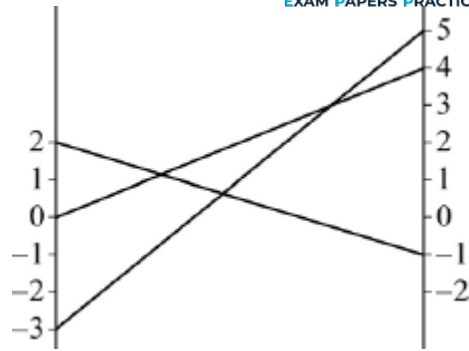
M1

C $-p + 2(1 - p)$

All correct

A1

[gives $4p, 8p - 3, 2 - 3p$]



Must have 3 lines

All correct with values shown

m1

A1

Max at $8p - 3 = 2 - 3p$

Identifies correct max from their graph

$$p = \frac{5}{11}$$

m1

A1

(J plays) E prob $\frac{5}{11}$, F prob $\frac{6}{11}$

A1

CSO

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(iii) Value of game = $\frac{7}{11}$

B1

1

[15]

Q5.

(a) For each pair of strategies

Roz gain + Colum gain = 0

E1 for general idea of

Roz gain + Colum gain = 0

E2,1

2

(b) Colum's max are -2, 3, -1

min (colum max) = -2

must see these values for E1

E1

$\Rightarrow$ play safe is C_1
 $C_1 \quad C_2 \quad C_3$

B1

2

(c) (i) Delete R_2 (PI by further work)
 $-2 \quad -6 \quad -1 \quad R_1$

M1

Since R_3 dominates R_2
 $-3 \quad 3 \quad -4 \quad R_3$

A1

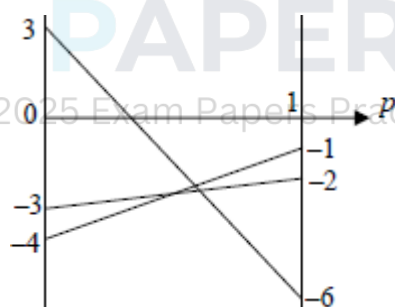
2

(ii) Let Roz play R_1 with prob p
 C_1 expected gain: $-2p - 3(1 - p) = p - 3$
 C_2 : $-6p + 3(1 - p) = 3 - 9p$
2 expressions unsimplified ft their matrix

M1

C_3 : $-p - 4(1 - p) = 3p - 4$
all correct

A1



plotting 3 expected gains for $0 \leq p \leq 1$

M1

correct gains plotted accurately

A1

Solving $p - 3 = 3 - 9p$
choosing highest point of 'their' region or correct pair solved

m1

$\Rightarrow 10p = 6$

$$p = \frac{3}{5}$$

A1

$\Rightarrow$ Roz plays R_1 with probability $\frac{3}{5}$ and

R_3 with probability $\frac{2}{5}$
must see R_1 and R_3

E1cao

7

[13]

Q6.

(a) (i)

Row min - 6, - 3, - 5, - 4

Max (row min) = - 3

Col max 5, 4, - 3

Min (col max) = - 3

*attempt to find maximin and minimax
 condone one slip in values*

M1

**all rows min and col max values correct
 and max (row min) = - 3 identified
 and min (col max) = - 3 identified**

A1

max (row min) = min (col max) = - 3

$\Rightarrow$ game has a stable solution

**full statement involving maximin and minimax and both
 values = - 3**

E1

3

(ii) Adam plays A_2 & Bill plays B_3

B1

1

(iii) Value of game for Bill is +3

B1

1

Examiners must use the correct symbol for marks carried forward, ie ringed totals with arrows through them.

- (b) (i) (Never play) C_2
correct strategy stated

C_2 dominated **by** C_1 ($-3 > -4$ and $2 > 1$)
and correct reason
condone $3 < 4$ and $-2 < -1$

B1

1

- (ii) $C_1 : 3p - 2(1 - p)$
either correct unsimplified

M1

$C_3 : -3p + 5(1 - p)$
both correct unsimplified
 $\{5p - 2, 5 - 8p\}$

A1

2

- (iii) $3p - 2(1 - p) = -3p + 5(1 - p)$
equating their 2 expressions

M1

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A1

2

- (iv) Value of game $= 5 \times \frac{7}{13} - 2$
or $5 - 8 \times \frac{7}{13}$

$$= \frac{9}{13}$$

B1

1

[11]

Q7.

- (a) (i) Row minima 2, -3 , x
Check for answers written on table

B1

1

- (ii) Column maxima 3, 6, 4

Check for answers written on table

B1

$$\text{Max (row min)} = 2$$

$$\text{Or } 2 \neq 3$$

$$\text{Min (col max)} = 3$$

Condone Best (worst) = 2 etc

Worst (best) = 3

M1

Since $2 \neq 3 \rightarrow$ no stable solution

Both lines and statement must score previous B1, B1

A1cso

3

(b)

$$\left. \begin{array}{l} x < 2, x + 3 < 6, 3 < 4 \\ \rightarrow R_1 \text{ dominates } R_3 \end{array} \right\} \text{ Either of these}$$

hence Rhona should not play R_3

B1

1

- (c) (i) Let Rhona play R_1 with prob p
and R_2 with prob $1 - p$

When C plays

$$C_1 : \text{exp value} = 2p + 3(1 - p)$$

$$= 3 - p$$

$$C_2 : 6p - 3(1 - p)$$

$$= -3 + 9p$$

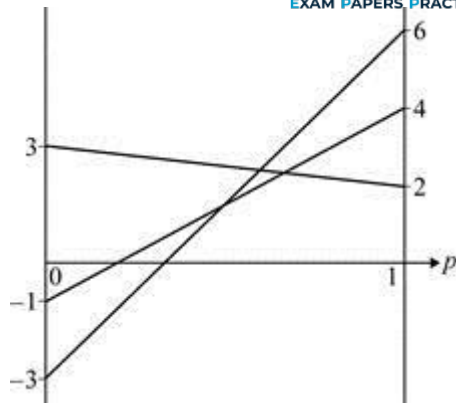
$$C_3 : 4p - (1 - p) = -1 + 5p$$

any two correct unsimplified

M1

all correct unsimplified

A1



drawing two of their expected values for $0 \leq p \leq 1$
both vertical axes using same scale
condone use of horizontal lines in paper

M1

all three correct lines
must see numbers on at least one vertical axis

A1

$$3 - p = -1 + 5p$$

choosing highest point of region

M1

$$\rightarrow p = \frac{2}{3}$$

A1

$\rightarrow$ Rhona plays R_1 $\frac{2}{3}$ of time

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and R_2 $\frac{1}{3}$ of time

ft their p

E1✓

7

$$\begin{aligned} \text{(ii) Value of game} &= 3 - \frac{2}{3} = \frac{7}{3} \\ \text{or} & -1 + \frac{10}{3} = \frac{7}{3} \end{aligned}$$

B1

1

[13]

Q8.

(a) Row minima are $-4, -3, -7$

both row minima **and** column maxima attempted
(condone 2 errors)

M1

Column maxima are $-3, 6, 8$
all values correct

A1

$\max(\text{row min}) = \min(\text{col max}) = -3$
condone arrows pointing to this element but
must **state** max (row min) and min (col max)
or equivalent

E1

Play-safe Tom II and Jerry A

B1

4

- (b) (i) Let Rohan play R_1 with prob p
 $\Rightarrow$ plays R_2 with prob $1 - p$

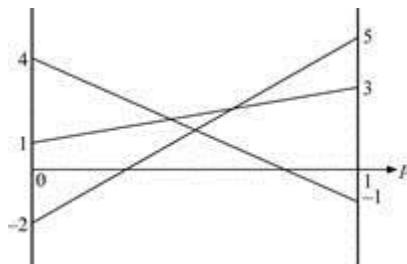
When Carla plays C_1 ,
Rohan's expected gain $= 3p + (1 - p)$
 $= 1 + 2p$

$C_2 : 5p + (-2)(1 - p) = 7p - 2$
at least 2 expected gains correct unsimplified

M1

$C_3 : -p + 4(1 - p) = 4 - 5p$
all 3 correct unsimplified

A1



at least 2 lines correct

M1

all lines correct for $0 \leq p \leq 1$ and values
at 0 and 1 clear

A1

$$7p - 2 = 4 - 5p$$

$$12p = 6$$

choosing highest point or using correct equation

M1

$$\Rightarrow p = \frac{1}{2} \Rightarrow$$

Rohan plays R_1 50% of the time and
 R_2 50% of the time

A1cso

$$\text{Value of game} = 7 \times \frac{1}{2} - 2 = \frac{3}{2} \quad \text{AG}$$

$$\text{or } 4 - \frac{5}{2} = \frac{3}{2} \quad \text{must see working}$$

B1

7

(ii) When Rohan plays R_1 , expected loss for Carla is $3p + 5q + (-1)(1 - p - q)$

and when Rohan plays R_2 , expected loss for Carla is $p + (-2)q + 4(1 - p - q)$

either expression correct unsimplified

M1

$$4p + 6q = \frac{3}{2} + 1$$

$$3p + 6q = 4 - \frac{3}{2}$$

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correct simultaneous equations unsimplified

A1

$$\Rightarrow p = 0, q = \frac{5}{12}$$

condone 0.42 or better

A1

$\Rightarrow$ Carla never plays C_1 ,

plays C_2 with prob $\frac{5}{12}$

and plays C_3 with prob $\frac{7}{12}$

Must have all 3 correct probabilities

Q9.

(a)

				Row min
				-2
				-3
				-5
Col max	4	0	-2	

Row minima and column maxima (all values)

B1

$$\text{Max (row min)} = -2$$

$$\text{Min (col max)} = -2$$

Both attempted or stated/indicated

M1

Since these are equal, there is a stable solution

Must have both values = -2 plus statement (withhold if max (min) and min (max) not stated)

A1

Ann plays A_1 and Bill plays B_3 for play-safe

E1

4

(b) (i) Let Russ play R_1 with probability p

And R_2 with probability $1 - p$

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$$C_1 : \text{expected gain } -4p + 2(1 - p) \\ (2 - 6p)$$

$$C_2 : 7p - (1 - p) = 8p - 1$$

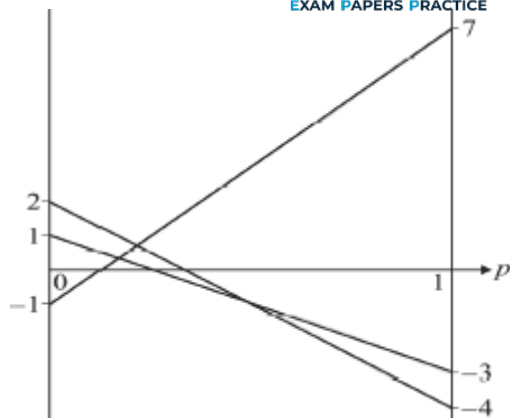
2 correct unsimplified

M1

$$C_3 : 3p + (1 - p) = 1 - 4p$$

All correct

A1



Plotting 3 expected gains for $0 \leq p \leq 1$

M1

Correct gains plotted accurately

A1

Solving $8p - 1 = 1 - 4p$

Choosing highest point of their region or correct

M1

$$\Rightarrow p = \frac{1}{6}$$

A1

$\Rightarrow$ Russ plays R_1 with probability $\frac{1}{6}$

and R_2 with prob $\frac{5}{6}$

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7

(ii) Value of game = $\frac{8}{6} - 1$

Or $1 - \frac{4}{6}$

$$= \frac{1}{3}$$

B1

1

[12]

Q10.

- (a) (i) Let Roger play R_1 with probability p and R_2 with probability $1 - p$

Expected gains:

$$C_1 : 7p - 2(1 - p) = 9p - 2$$

$$C_2 : 3p - (1 - p) = 4p - 1$$

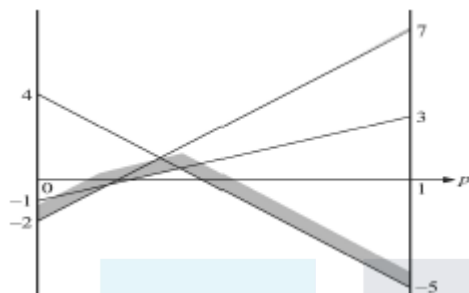
$$C_3 : -5p + 4(1 - p) = 4 - 9p$$

one correct unsimplified

M1

all correct unsimplified

A1



2 of their lines drawn correctly

M1

*all correct and accurate for $0 \leq p \leq 1$
 Condone lines not quite to $p = 1$ if using
 "accurate" intersection points on p -axis
 i.e. $\frac{2}{9} < \frac{1}{4}$ and $\frac{4}{9} \approx \text{twice } \frac{2}{9}$*

A1

C_2 and C_3 lines give optimum

$$4p - 1 = 4 - 9p$$

ft their max point of region

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$$p = \frac{5}{13}$$

*Condone 0.385 or 0.3846(15...) must be
 correct rounding if 3sf used*

A1

Roger plays

$$R_1 \frac{5}{13} \text{ of time and } R_2 \frac{8}{13} \text{ of time}$$

CAO

E1

7

(ii) Value of game $= 4 \times \frac{5}{13} - 1 = \frac{7}{13}$

$$AG \quad \text{or} \quad \left(4 - 9 \times \frac{5}{13}\right) = \frac{7}{13}$$

must see correct calculation

B1

1

- (b) Let Corrie play C_1 with prob p , C_2 with prob q , C_3 with prob $1 - p - q$
 $R_1 : 7p + 3q - 5(1 - p - q)$
 $R_2 : -2p - q + 4(1 - p - q)$
 any correct expression

M1

$$\Rightarrow 12p + 8q = 5 \frac{7}{13}$$

$$6p + 5q = 3 \frac{6}{13}$$

either equation correctly with coefficients of p and q correctly simplified

A1

$$\Rightarrow \left. \begin{array}{l} q = \frac{9}{13} \\ p = 0 \end{array} \right\}$$

may reason that $p(C_1) = 0$ from part (a) E1 with M1, A1, A1, E1 from 2×2 equations

$$3r - 5s = \frac{7}{13}$$

$$-r + 4s = \frac{7}{13}$$

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m1A1CSO

$\Rightarrow$ Optimal mixed strategy is
 C_1 with prob 0

C_2 with prob $\frac{9}{13}$

Condone 0.692

C_3 with prob $\frac{4}{13}$

CAO & 0.308

E1

5

[13]

Q11.

(a)

Row minima

-7
 -1
 -3

Column max 6 8 -1

Max (row min) = -1

*Attempting Row Min & Col Max
 or Maxmin and Minmax*

M1

Min (col max) = -1

*All values correct and shown with
 correct words*

A1

Since these values are equal the game has a stable solution

Must both be -1 and have statement

E1

Raj plays II, Cal plays Z

B1

4

(b) (i) $C_1 : 5p - 2(1 - p)$
 $7p - 2$

B1

$C_2 : xp + 4(1 - p)$

B1

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2

(ii) Value of game = $\frac{8}{3}$
 $\Rightarrow 5p - 2(1 - p) = \frac{8}{3}$

$$\text{Their expected gain } (C_1) = \frac{8}{3}$$

M1

$\Rightarrow p = \frac{2}{3}$

A1

$xp + 4(1 - p) = \frac{8}{3}$

$\frac{8}{3}$
 Their C_2 gain = $\frac{8}{3}$ (must involve x)

M1

$$\Rightarrow \frac{2}{3}x + \frac{4}{3} = \frac{8}{3}$$

$$\Rightarrow x = 2$$

A1

4

[10]

Q12.

- (a) (For each outcome)
 Rowena's gain + Colin's gain = 0
One player's loss is other's gain

E1

1

- (b) (Column maxima 2, 5, 4)
 $\Rightarrow \min(\text{colmax}) = 2$ (OE but strict)
Withhold E mark if any value incorrect; accept column minimax = 2

E1

$\Rightarrow$ Colin's play-safe strategy is C_1

B1

2

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- (c) R_3 is dominated by R_1
 $-5 < -4$; $4 < 5$ and $3 < 4$
E0 if R_2 mentioned as well

E1

1

- (d) Let Rowena play R_1 with prob p
 and R_2 with prob $1 - p$
 Expected gain when Colin plays
 $C_1 : -4p + 2(1 - p) = 2 - 6p$
 $C_2 : 5p - 3(1 - p) = -3 + 8p$
 $C_3 : 4p - (1 - p) = -1 + 5p$
attempt at least 2 with one correct

M1

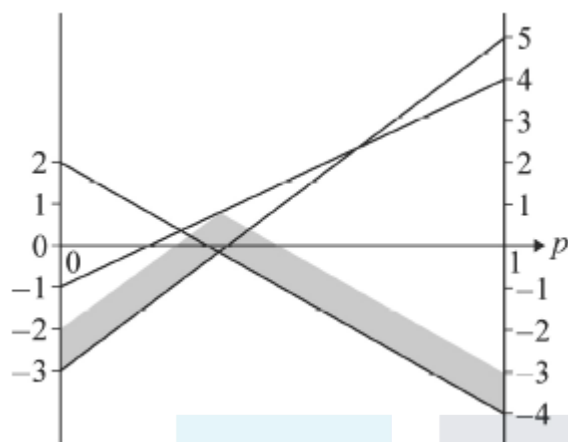
all 3 correct unsimplified

A1

Plot expected gains against p for $0 \leq p \leq 1$

All 3 drawn ft their exp gains

M1



correct

A1

$$\Rightarrow 2 - 6p = -3 + 8p$$

Using "correct" equation
Choosing highest point of region

M1

$$\Rightarrow p = \frac{5}{14}$$

A1

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Therefore Rowena plays R_1 with prob $\frac{5}{14}$

and R_2 with prob $\frac{9}{14}$

ft their p

E1ft

7

[11]

Q13.

- (a) Rob's gain = Con's loss
(at each entry of matrix)

Zero-sum explained

Rob's winnings + Con's winnings = 0
(for every pair of strategies)

E1

1

(b)

	-2	5	3	min
	3	-3	-1	-2
	-3	3	2	-3
max	3	5	3	

min of rows and max of columns
All values correct (seen)

B1

$$\left. \begin{array}{l} \text{maximin} = -2 \\ \text{minimax} = 3 \end{array} \right\} \text{either correct}$$

M1

$-2 \neq 3$
 $\Rightarrow$ no stable solution

E1

3

- (c) R_3 dominated by R_1
 $(-3, 3, 2) < (-2, 5, 3)$ so never play R_3

E1

1

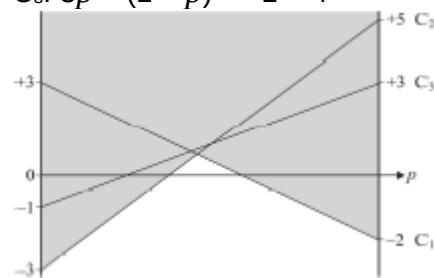
- (d) (i) Choose R_1 with probability p
 and R_2 with probability $1 - p$

Expected gain when C plays:

$$C_1: -2p + 3(1 - p) = 3 - 5p$$

$$C_2: 5p - 3(1 - p) = 8p - 3$$

$$C_3: 3p - (1 - p) = -1 + 4$$



$$3 - 5p = 8p - 3$$

Attempt at one expression

M1

All correct unsimplified

A1

Plotting expected gain for $0 \leq p \leq 1$

M1

Correct with values at $p = 0$ and $p = 1$ clear

A1

Choosing C_1 and C_2 intersection or their highest point

M1

$$\Rightarrow p = \frac{6}{13}$$

A1

Play R_1 with probability $\frac{6}{13}$
and R_2 with probability $\frac{7}{13}$
FT their p (statement needed)

E1ft

7

(ii) Value of game $= 3 - \frac{30}{13}$
Or $\frac{48}{13} - 3$
 $= \frac{9}{13}$

B1

1

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[13]

Q14.

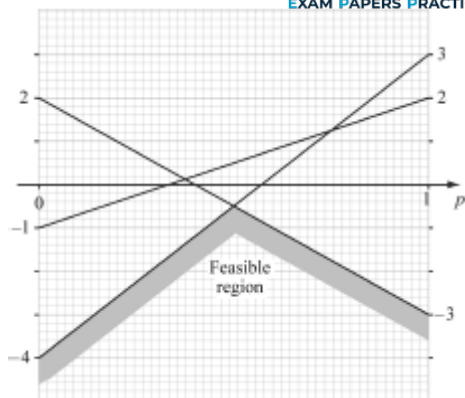
- (a) (i) Roseanne plays R_1 with prob p
Expected value when Collette plays
 $C_1 : -3p + 2(1 - p) = 2 - 5p$
 $C_2 : 2p - (1 - p) = 3p - 1$
 $C_3 : 3p - 4(1 - p) = 7p - 4$

One correct unsimplified

M1

All correct unsimplified

A1



drawing 'their' lines (2 'correct' ft)

M1

correct with values clear at $p = 0$ and $p = 1$

A1

Solving $2 - 5p = 7p - 4$

$$6 = 12p$$

their highest point

M1

$$\Rightarrow p = \frac{1}{2}$$

A1

SC B1 if $p = \frac{1}{2}$ found from graph

Strategy is to play R_1 for 50% of time

E1ft

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7

(ii) Value $= 2 - 5\left(\frac{1}{2}\right)$ or $7\left(\frac{1}{2}\right) - 4 = -\frac{1}{2}$

AG CSO

$$p = \frac{1}{2} \text{ and both expressions correct}$$

B1

1

- (b) (i) Let Collette play C_1 with prob p
and C_2 with prob q
 $\Rightarrow C_3$ with prob $1 - p - q$

B1

1

$$(ii) \quad -3p + 2q + 3(1 - p - q) = -\frac{1}{2}$$

$$2p - q - 4(1 - p - q) = -\frac{1}{2}$$

Either equation LHS correct
Condone $(1 - p + q)$ used

M1

$$\Rightarrow 6p + q = 3\frac{1}{2}$$

$$6p + 3q = 3\frac{1}{2}$$

Either equation
correct and simplified p & q coefficients

A1

$$\Rightarrow p = \frac{7}{12}$$

$$q = 0$$

CSO

A1

$\Rightarrow$ Collette plays C_1 with prob $\frac{7}{12}$,
 (never plays C_2),
 and plays C_3 with prob $\frac{5}{12}$

Must have statement with C_1 & C_3
correct only

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E1

4

[13]

Q15.

(a) (i)

Row min

-4
 -2
 -1

Col max 5 -1 3

Attempt at row minimum and column maximum

M1

all figures correct

A1

$\min(\text{col max}) = \max(\text{row min})$
 $\Rightarrow$ stable solution

E1

3

(ii) Ros plays III and Col plays Y

B1

value of game = -1

B1

2

(b) (i) Ros plays R_1 with probability p and R_2 with probability $1 - p$

Expected gains when Col plays:

$$C_1 : 3p - 2(1 - p) = 5p - 2$$

$$C_2 : 2p - (1 - p) = 3p - 1$$

$$C_3 : p + 2(1 - p) = 2 - p$$

attempt at least 2

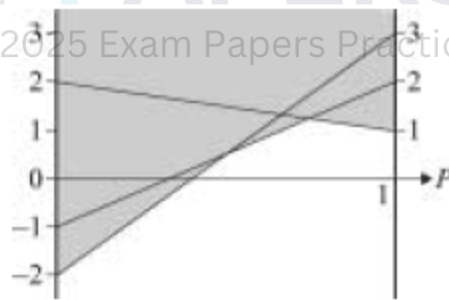
M1

correct unsimplified

A1

Plot expected gains against p for $0 \leq p \leq 1$

M1



correct (must see 0 or 1 on P axis, or implied by their numbers)

A0 if not possible to see highest point of region being correct

A1

Choose highest point of region below

lines $\Rightarrow 3p - 1 = 2 - p$

must be this pair of lines or their highest point

M1

leading to $p = \frac{3}{4}$

A1

Therefore Ros plays R_1 with prob $\frac{3}{4}$
and plays R_2 with prob $\frac{1}{4}$
ft their p from any lines

B1ft

7

(ii) Value of game $= 3 \times \frac{3}{4} - 1$
or $\left(2 - \frac{3}{4}\right) = 1\frac{1}{4}$

B1

1

[13]

Q16.

- (a) (i) $\text{Min } R_1 (5, 2, -1) = -1$
 $\text{Min } R_2 (-3, -1, 5) = -3$
 $\text{Min } R_3 (4, 1, -2) = -2$

E1

$\text{Max min} = -1$
 $\Rightarrow$ Play safe strategy R_1

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B1

2

- (ii) $\text{Max } C_1 = 5; \text{ max } C_2 = 2; \text{ max } C_3 = 5$
 $\text{Min } (5, 2, 5) = 2$

M1

$2 \neq 1 \Rightarrow$ no stable solution

A1

2

- (b) $R_3 (4, 1, -2) < R_1 (5, 2, -1)$

E1

1

- (c) (i) C_1 played, expected gain for Rose:
 $5p + -3(1 - p)$
Any correct expected gain unsimplified

M1

$$= 8p - 3$$

One correct simplified

A1

$$C_2 \div p - (1 - p) = 3p - 1$$

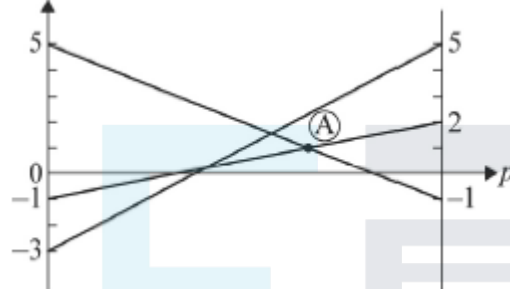
$$C_3 \div -p + 5(1 - p) = 5 - 6p$$

All correct simplified

A1

3

(ii) Expected gain



Plotting at least 2 lines

M1

All correct with values at $p = 0$ and $p = 1$ indicated

A1

2

(iii) Choosing A – highest point in feasible region

$$\Rightarrow 3p - 1 = 5 - 6p$$

$$9p = 6$$

Solving this equation

M1

$$\Rightarrow p = \frac{2}{3}$$

CSO

A1

$$\Rightarrow \text{Rose plays } R_1 \frac{2}{3} \text{ of time}$$

$$\text{and } R_2 \frac{1}{3} \text{ of time}$$

E1ft

3

(iv) Value of game = $3 \times \frac{2}{3} - 1 = 1$
 Or $5 - 4 = 1$

B1

1

[14]

Q17.

(a) $(-2, 2, 4) < (2, 4, 5)$

So S_1 dominated by S_2

E1

$$\begin{pmatrix} 4 \\ 5 \\ 2 \end{pmatrix} > \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix}$$

note > sign

So S_3 dominated by C_2

E1

2

(b) $C_1 \quad C_2$

$$\begin{matrix} S_2 \\ S_3 \end{matrix} \begin{bmatrix} 2 & 4 \\ 5 & 1 \end{bmatrix}$$

2×2 game now

Minimum of rows $(2, 4) = 2$

correct method for either S or C

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M1

Minimum of $(5, 1) = 1$

Choose maximum = 2

play safe for Sam is S_2

A1

Max of column 1 = $\max(2, 5) = 5$

Max of column 2 = $\max(4, 1) = 4$

Choose minimum = 4

play safe for computer is C_2

AI

Since $2 \neq 4 \Rightarrow$ not stable solution

E1

- (c) (i) Computer picks C_1

$$\text{Expected game} = 2p + 5(1 - p)$$

M1

$$= 5 - 3p$$

A1

Computer picks C_2

$$\text{Expected gain} = 4p + (1 - p)$$

$$= 1 + 3p$$

A1

3

- (ii) Best mixed strategy

$$5 - 3p = 1 + 3p$$

M1

$$\Rightarrow p = \frac{2}{3}$$

M1 A1

2

- (iii) Expected points gain

$$= 5 - 3 \times \left(\frac{2}{3}\right)$$

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$$\text{Or } 1 + 3 \left(\frac{2}{3}\right)$$

$$= 3$$

B1

1

[12]

Q18.

- (a) Gain for Rowan + gain for Colleen in each strategy = 0

Gain for one = loss of other

E1

1

- (b)

-3	-4	1	-4
1	5	-1	-1
-2	-3	4	-3
Max	1	5	4

$\left\{ \begin{array}{l} \text{minimum of rows \& max of columns} \\ \text{or} \\ \text{maximum of minima or minimax} \end{array} \right.$

M1

All values correct (seen) or words maximin and minimax highlighted

A1

$1 \neq -1 \Rightarrow$ no stable solution

E1

3

- (c) R_3 dominates R_1
 $(-3, -4, 1) < (-2, -3, 4)$ so never play R_1

E1

1

- (d) (i) R chooses R_2 with prob p
 Attempt at one expression

M1

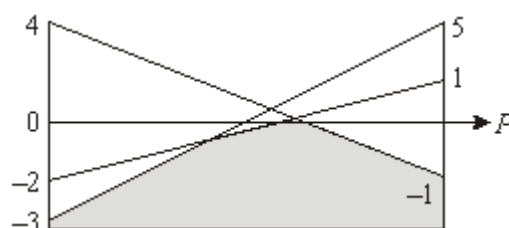
$\Rightarrow$ choose R_3 with prob $1 - p$
 $\Rightarrow$ expected gain when C plays
 $C_1 : p - 2(1 - p) = 3p - 2$
 $C_2 : 5p - 3(1 - p) = 8p - 3$
 $C_3 : -p + 4(1 - p) = 4 - 5p$

All correct unsimplified

A1

Plot expected gains for $0 \leq p \leq 1$

M1



Condone mirror image

A1

Choosing their "highest" point
Any 2 lines

M1

$$C_1 \text{ \& } C_3 \text{ intersect } \Rightarrow 3p - 2 = 4 - 5p$$

$$\Rightarrow p = \frac{3}{4}$$

A1

$\Rightarrow$ play R_2 with prob $\frac{3}{4}$
and R_3 with prob $\frac{1}{4}$

Statement of strategy

E1ft

7

(ii) Value of game is $3 \times \frac{3}{4} - 2 = \frac{1}{4}$
CSO or equivalent, e.g. 0.25

B1

1

[13]

EXAM PAPERS PRACTICE

Q19.

(a)

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				Min
	4	3	5	3*
	-1	5	-2	-2
	1	2	3	1
Max	4*	5	5	

Either marginal row or column

M1

all values correct

A1

Since $3 \neq 4 \Rightarrow$ no stable solution

A1

3

- (b) P_1 dominates P_3 ; $(1,2,3) < (4,3,5)$
 So it is unwise to play P_3

E1

1

- (c) P chooses P_1 with probability p
 So chooses P_2 with probability $1 - p$

Expected gains when Q plays

Attempt at at least 2

M1

$$Q_1: 4p - (1 - p) = 5p - 1$$

$$Q_2: 3p + 5(1 - p) = 5 - 2p$$

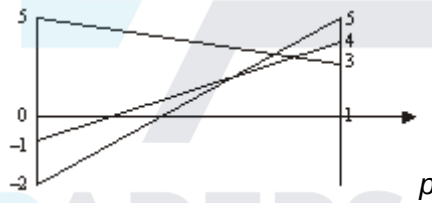
$$Q_3: 5p - 2(1 - p) = 7p - 2$$

All 3 correct (simplified)

A1

Plot expected gains against p for
 $0 \leq p \leq 1$

M1



A1

Choose highest point of region below lines

$$5p - 1 = 5 - 2p$$

M1

$$\text{leading to } p = \frac{6}{7}$$

A1

Therefore P plays P_1 with probability $\frac{6}{7}$
 and plays P_2 with probability $\frac{1}{7}$

B1ft

7

[11]

Q20.

- (a) For A
 Row I > Row III
 Or row II > Row III

E1

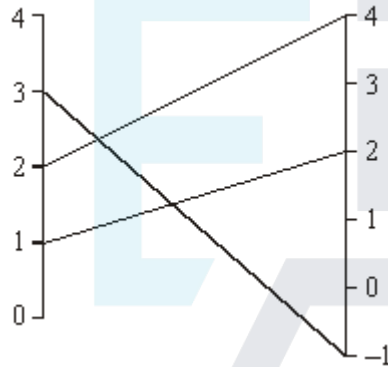
1

- (b) (i) A plays I prob p
 II prob $1 - p$

M1

$$\begin{aligned}
 \text{Returns are } 4p + 2(1 - p) &= 2 + 2p \\
 -p + 3(1 - p) &= 3 - 4p \\
 2p + (1 - p) &= 1 + p
 \end{aligned}$$

A2,1,0



M1A1

$$3 - 4p = 1 + p \Rightarrow p = \frac{2}{5}$$

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A1

$$\text{Hence I} - \frac{2}{5} \quad \text{II} - \frac{3}{5}$$

B1

7

(ii) $\therefore V = 1 + \frac{2}{5} = \frac{7}{5}$

B1

1

- (c) For B play I prob p
 II prob q
 $p = 0$

Or

III prob $1 - p - q$

Returns are

M1

$$-q + 2(1 - q) \left(= \frac{7}{5} \right)$$

$$4p - q + 2(1 - p - q) \left(= \frac{7}{5} \right)$$

M1A1

$$2p + 3q + (1 - p - q) \left(= \frac{7}{5} \right)$$

A1

$$3q = \frac{3}{5}$$

$$\Rightarrow 2p - 3q = -\frac{3}{5}$$

$$p + 2q = \frac{2}{5}$$

M1

$$q = \frac{1}{5}$$

$$\therefore q = \frac{1}{5}, p = 0$$

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A1

4

Or

$$p = 0$$

M1

$$-1q + 2(1 - q) = 3q + 1 - q$$

M1A1

$$2 - 3q = 2q + 1$$

$$1 = 5q$$

$$\frac{1}{5} = q$$

A1

[13]

Q21.

(a) For A, $III > I$

E1

For B, $I > III$

or $II > III$

E1

2

(b)

	I	II
p	3	1
1 - p	2	3

sight of p, $1 - p$

M1

If Ben plays I:

return $3p + 2(1 - p) = p + 2$

$kp + c(1 - p)$; their $(1 - p)$

M1

If Ben plays II:

return $p + 3(1 - p) = 3 - 2p$

both correct

A1

$$\therefore p + 2 = 3 - 2p$$

$$3p = 1$$

$$p = \frac{1}{3} \quad \left(1 - p = \frac{2}{3}\right)$$

A1

Value of games $1 = 2 \frac{1}{3}$

B1F

	q	1 - q
I	3	1
II	2	3

sight of q, $1 - q$

M1

If Arnie plays I:

return $3q + 1(1 - q) = 2q + 1$

$kq + c(1 - q)$; their $(1 - q)$

M1

If Arnie plays II:

return $2q + 3(1 - q) = -q + 3$

$$\therefore 2q + 1 = -q + 3$$

A1

$$q = \frac{2}{3} \quad \left(1 - q = \frac{1}{3} \right)$$

$$\text{or } 2q + 1 = 2 \frac{1}{3}$$

$$q = \frac{2}{3}$$

A1

9

[11]

Q22.

(a) (i) Row min $(-2, 2, -3)$

M1

Max 2
 Column max $(3, 5, 2, 6)$
 Min 2

A1

$2 = 2$, therefore stable solution

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E1

3

(ii) 1 saddle point at $(2, 3)$

B1

1

(b) Colin plays I with prob p , II with $(1 - p)$

M1

Return $p(x + 2) + 3(1 - p)$

A1

$$p(x - 1) + 5(1 - p)$$

A1

$$p(x-1) + 3 = \frac{19}{5}$$

equating to value

M1

$$p(x-6) + 5 = \frac{19}{5}$$

$$p(x-1) = \frac{4}{5}$$

or solving for p first

A1

$$p(x-6) = -\frac{6}{5}$$

A1

$$\frac{4}{5(x-1)} = \frac{-6}{5(x-6)}$$

M1

$$4x - 24 = -6x + 6$$

$$x = 3$$

CAO

A1

EXAM PAPERS PRACTICE ⁸ [12]

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Q23.

(a) Min rows $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ Max = 2

M1

Max cols Min = 4

(7, 4, 6)

A1

$2 \neq 4$

E1

3

(b) III > II

B1

$$\begin{bmatrix} 1 & 3 & 6 \\ 7 & 4 & 2 \end{bmatrix}$$

B1

2

- (c) Let Bev play with
 $p, q, 1 - p - q$

M1

$$p + 3q + 6(1 - p - q) = 3\frac{3}{5}$$

A1 without $\left(= 3\frac{3}{5}\right)$

A1

$$7p + 4q + 2(1 - p - q) = 3\frac{3}{5}$$

A1

$$-5p - 3q = -2\frac{2}{5}$$

$$5p + 2q = 1\frac{3}{5}$$

$$\Rightarrow q = \frac{4}{5}, p = 0$$

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A1

Plays I prob 0

II prob $\frac{4}{5}$

III prob $\frac{1}{5}$

All three

B1

5

[10]

Q24.

- (a) Not drop shot
as $R1 > R2$
or $R3 > R2$

B1

E1

2

- (b) D chooses B p
 D chooses L $1 - p$

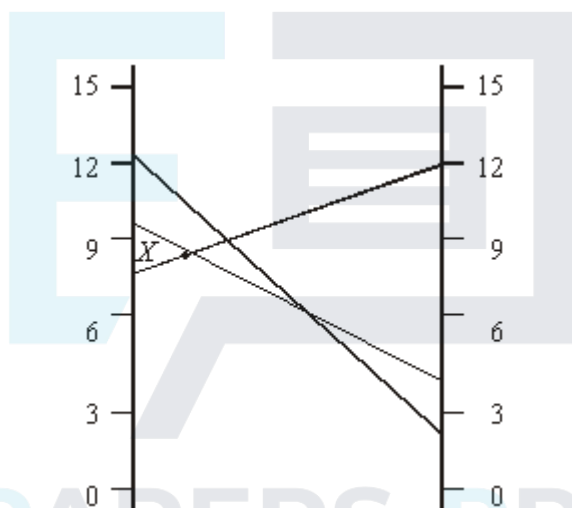
M1

$$\begin{aligned}
 \text{If T chooses B} &\Rightarrow 12p + 8(1 - p) \\
 \text{D} &\Rightarrow 5p + 10(1 - p) \\
 \text{L} &\Rightarrow 2p + 13(1 - p)
 \end{aligned}$$

A1

$$\begin{array}{ll}
 B = 4p + 8 & (8, 12) \\
 D = 10 - 5p & (10, 5) \\
 L = 13 - 11p & (13, 2)
 \end{array}$$

A1



M1 A2

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Max at X

B1

$$4p + 8 = 10 - 5p$$

$$p = \frac{2}{9}$$

B1

David should play
 Boast 2 shots to every Lob 7 shots (2 : 7)
Both

B1

8

(c) Value = $\frac{8}{9} + 8$
 $= 8\frac{8}{9}$

B1F

1

[11]

Q25.

(a)

				Min
	5	1	3	1
	2	5	4	2
	4	-1	2	-1
Max	5	5	4	

$2 \neq 4 \Rightarrow$ no stable solution

A1

2

(b) $(5 \ 1 \ 3) > (4 \ -1 \ 2)$

E1

1

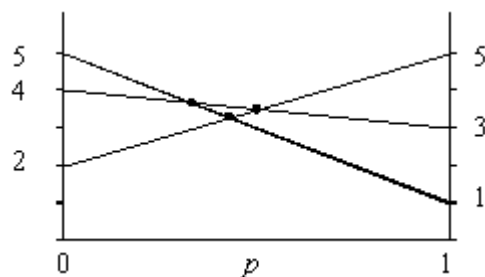
(c) (i) A chooses 1 p
 chooses 2 $1-p$

M1

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$\therefore$ gain $5p + 2(1-p) = 3p + 2$
 $1p + 5(1-p) = 5 - 4p$
 $3p + 4(1-p) = 4 - p$

A1



or $3p + 2 = 5 - 4p \quad p = \frac{3}{7}$

M1

$$3p + 2 = 4 - p \quad p = \frac{1}{2}$$

A1

$$5 - 4p = 4 - p \quad p = \frac{1}{3}$$

A1

Therefore $p = \frac{3}{7}$
choosing the middle value

B1F

Therefore A plays 1 with $\frac{3}{7}$
2 with $\frac{4}{7}$

B1F

7

(ii) Therefore the value is $3 \times \frac{3}{7} + 2 = \frac{23}{7}$

B1

1

[11]

Q26.

(a) For A $(5 \ 2 \ 7) > (4 \ 1 \ 5)$

either comparison

M1

For B $\begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} < \begin{pmatrix} 7 \\ 5 \\ 6 \end{pmatrix}$

A1

A1

3

(b) A chooses I p
A chooses II $(1 - p)$
If B chooses I, gain = $5p + 3(1 - p)$
If B chooses II, gain = $2p + 5(1 - p)$
 $5p + 3(1 - p) = 2p + 5(1 - p)$
or graphical

M1

A1

A1

M1

$$\Rightarrow p = \frac{2}{5}$$

A1

B chooses I q
B chooses II $(1 - q)$

M1

$$5q + 2(1 - q) = 3q + 5(1 - q)$$

A1

$$\Rightarrow q = \frac{3}{5}$$

B1

$$\text{Value} = 2p + 3 = 3\frac{4}{5}$$

s.c. A and B interchanged maximum 7 marks
(-2 from solution wherever it stops)

B1

EXAM PAPERS PRACTICE 9 [12]

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Examiner reports

Q3.

This question was demanding, but it was encouraging to see many fully correct solutions. For part (a), students needed to state that strategy B was dominated by strategy C, and not merely state that ‘other strategies are better’. Some students lost marks in part (b) by not drawing an accurate graph. This was essential in this question as the intersections of the lines were close together. Although it has been stated in previous reports on the examination for this unit, it is essential that an accurate graph with a scale on the vertical lines is used. A number of students failed to state the value of the game. The majority of students used probabilities of p , q and $(1 - p - q)$ in their attempt at part (c). Although many students obtained two correct expressions, a significant number failed to equate these expressions to the value of the game. A minority of students stated that the ‘probability of D was 0’, following from the previous part. These students almost always scored full marks.

Q4.

In parts (a) and (b), the majority of students had an idea of the requirements of this question, but full marks were rare. It is essential that students give full and clear statements to convince an examiner; simply writing down 6 numbers, circling two of the numbers, then stating ‘ $1 \neq -2$ ’ will not score full marks. Part (c) was well answered, although some students lost the mark through a lack of clarity in their statement. Part (d)(i) was poorly answered. Although students realized that the question was concerned with changing signs, the explanations given were poor and showed a lack of understanding of the problem. Many students scored full marks on the final 2 parts. Errors occurred when students failed to draw accurate graphs and consequently failed to identify the correct maximum point.

Q5.

Part (a) In most cases the explanation of a “zero-sum game” was inadequate for full marks; most candidates scored a single mark for a statement such as “one player’s gain is the other one’s loss”.

Part (b) The values of Colum’s column maxima needed to be calculated along with a statement that the minimum of these maxima was -2 in order to give an adequate reason for C_1 being Colum’s play safe strategy.

Part (c)(i) Most candidates gave a suitable reason based on domination as to why row 2 needed to be deleted; it was not enough to say that “other strategies are better than R_2 ”. Some deleted a column and were unable to make much progress with the rest of the question.

Part (c)(ii) Most candidates were able to find the optimal mixed strategy for Roz. Some lost marks for not drawing accurate lines showing expected gains for $0 \leq p \leq 1$; the intersection points of the three lines were very close together and so an accurate graph was essential here. Although it was acceptable to use the horizontal lines printed in the answer booklet, at least one of these vertical lines needed to have a clear scale so that the accuracy of the process of finding the highest point in the region could be checked by the examiner. Some omitted to say that Roz should play R_3 with probability 0.4 when stating the optimal strategy.

Q6.

- (a) (i) Explanations rarely scored full marks. In order to show that the game has a stable solution, it was expected that the maximum values in the columns would be indicated before finding the minimum value of these maxima. The maximum of the row minima also needed to be found. Some statement should then have been made indicating that these two values were equal and hence that the game has a stable solution. Quite a few students felt that all they needed to do was to draw arrows pointing to -3 when more detail was required. A few chose to solve the problem using an argument based on dominance, but the explanations were rarely adequate to earn more than a single method mark.
- (ii) It was encouraging to see many students correctly identify the play-safe strategy for each player.
- (iii) Only the better students stated the correct value of the game for **Bill**, despite the bold type; as you might expect weaker students gave the value as -3 , and others gave the value as 0 , possibly confused by the term “zero-sum game”.
- (b) *A large number of students were not prepared for this part of the question.*
- (i) A large number deleted the first column and scored only a couple of method marks for the rest of this question. Some who realised that the computer should not play C_2 spoiled their answers by writing such things as “each of the other strategies is better”.
- (ii) The better students managed to use the values in the table correctly to find correct expressions for the expected gains for Roza for the two remaining strategies. A few ignored the dominance from part(b)(i) and attempted to solve the problem graphically using three lines; some successfully, others not.
- (iii) Some poor algebra, usually when removing brackets, prevented some from finding the correct value of p , but most marks were lost for having the incorrect pair of simultaneous equations.
- (iv) Those who scored full marks in parts (b)(ii) and (iii) were usually able to obtain the correct value of the game for Roza.

Q7.

In part (a)(i) the entry “ x ” caused problems to several candidates. Credit was generously given to those who wrote the correct three minima at the ends of the rows in the pay-off matrix, when they were perhaps unaware of what was required in the request made in this part.

In part (a)(ii) explanations rarely scored full marks. In order to show that the game did not have a stable solution, it was expected that the maximum values in the columns would be indicated before finding the minimum value of these maxima. The maximum of the row minima from part (a)(i) also needed to be found. Some statement should then have been made indicating that these two values were not equal and hence that the game did not have a stable solution.

In part (b)(i) most candidates were able to find the optimal mixed strategy for Rhona. Some lost marks for not drawing accurate graphs of expected values for $0 \leq p \leq 1$. Although it is acceptable to use the horizontal lines printed in the answer booklet, at least one of these vertical lines should have a clear scale so that the accuracy of the process of

finding the highest point in the region can be checked by the examiner.

In (b)(ii) those finding the correct optimal strategy in part (b)(i) were usually able to find the correct value of the game.

Q8.

- (a) Explanations rarely scored full marks. In order to show that the game has a stable solution, it was expected that the maximum values in the columns would be indicated before finding the minimum value of these maxima. The maximum of the row minima also needed to be found. Some statement should then have been made indicating that these two values were equal and hence that the game has a stable solution. Quite a few candidates felt that all they needed to do was to draw arrows pointing to -3 when an explicit statement was required regarding the play-safe strategies for each player.
- (b) (i) Most candidates were able to find the optimal mixed strategy for Rohan. Very few made a statement such as “Let Rohan play R_1 with probability p ”; usually the letter p was introduced with no explanation. Some lost marks for not drawing accurate graphs of expected values for $0 \leq p \leq 1$. Although it is acceptable to use the horizontal lines printed in the answer book, at least one of these vertical lines should have a clear scale so that the accuracy of the process of finding the highest point in the region can be checked by the examiner. Those finding the correct optimal strategy were usually able to find the correct value of the game, but for full marks it was necessary to make a statement regarding the frequencies of playing the two strategies.
- (ii) A large number of candidates were not prepared for this part of the question. Despite the wording in the question, very few made the correct start using the **three** probabilities p , q and $1 - p - q$ in order to obtain a pair of simultaneous equations in p and q . Those who made a correct start usually completed and found the optimal mixed strategy for Carla; some candidates, however, did obtain values for p and q which were either negative or more than 1.

Q9.

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Some candidates argued correctly, in part (a), that row 2 dominated row 3 but then incorrectly that one column dominated another in an attempt to reduce the matrix to a 2 by 2. In order to show that the game had a stable solution, it was expected that the minimum values in the rows and maximum values in the columns would be indicated before finding the maximum of the minima and the minimum of the maxima. Some statement should then have been made indicating that these two values were equal and hence that the game has a stable solution. The actual play-safe strategies for Ann and Bill were sometimes stated incorrectly even when the correct minimax and maximin values had been identified in the table.

The method of finding the optimal mixed strategy in part (b) seemed to have been clearly understood by most candidates. Sometimes the graphs were inaccurate and a mark was lost.

Other good solutions were sometimes spoiled by not stating how often Russ should play each of the strategies R_1 and R_2 . Those who found the correct probabilities in part (b)(i) were usually able to find the value of the game in part (b)(ii), but some substituted into the wrong expression for expected gain.

Q10.

Part (a)(i) Despite the standard nature of this question, it was surprising to see several candidates who were unaware of how to find the optimal mixed strategy. A good sketch showing the feasible region was expected, with the highest point of the feasible region being selected in order to find the probability of playing the various rows. Many gave no reason for introducing a probability p and failed to describe the actual mixed strategy when the value of p had been found.

Part (a)(ii) The printed answer helped many candidates to check their previous results when finding the value of the game. Quite a few, who had previously found that $p = 5/13$, wrote that “the value of the game was $1 - p = 7/13$ ”

Part (b) A large number of candidates made no attempt at this part of the question. Others who only partly remembered the technique wrote down expressions such as $7p + 3(1 - p) - 5(1 - p - q)$ and scored no marks. A few reasoned correctly using the result from part (a) that C_1 should never be played and reduced their matrix to a simple 2 by 2 game and then solved a pair of simple equations or a single equation such as $3p - 5(1 - p) = 7/13$ which scored full marks, provided things had been clearly explained. Although it is not in the specification, one or two candidates used a Linear Programming Simplex approach with little explanation and no solution seen was deemed worthy of any marks.

Q11.

In part (a), in order to show that the game had a stable solution, it was expected that the minimum values in the rows and maximum values in the columns would be indicated before finding the maximum of the minima and the minimum of the maxima. Some statement should then have been made indicating that these two values were equal and hence the game has a stable solution.

Poor algebraic skills in part (b), together with the omission of brackets, caused many candidates to write down expressions which were often simplified incorrectly. Many failed to make progress because they failed to equate each of the expressions found in part (b)(i) to $8/3$; others obtained nonsensical answers such as $p = 4/3$ because of earlier errors in their equations. It was surprising how many took more than a page of working to solve these two simple linear equations.

Q12.

In part (a), few candidates mentioned that, **for each outcome**, one player's gain plus the other player's gain is zero in a zero-sum game.

In part (b), most responses indicated that C_1 was Colin's play-safe strategy. However, the accompanying explanations were often poor with many producing the standard table indicating that there was no stable solution when this was not required. Better candidates gave the column maxima, namely 2, 5 and 4, together with a comment about 2 being the minimum of these column maxima.

In part (c), the idea that R_3 was dominated by R_1 was well understood, but some candidates spoiled their answer by suggesting that strategy R_3 was worse than both R_1 and R_2 .

Despite the standard nature of this question, it was surprising to see many candidates unaware of how to find the optimal mixed strategy in part (d). A good sketch showing the feasible region was expected with the highest point of the feasible region being selected in

order to find the probability of playing the various rows. Many gave no reason for introducing a probability p nor explained what the actual mixed strategy was when the value of p had been found. Many candidates calculated the value of the game, which was not required

Q13.

Part (a) The term “zero-sum” was generally understood, but candidates should have indicated that the sum of Rob’s winnings and Con’s winnings was equal to zero *for every pair of strategies*.

Part (b) In order to show that the game has a stable solution, it was expected that the minimum values in the rows and the maximum values in the columns would be indicated before finding the maximum of the minima and the minimum of the maxima. Some statement should then have been made indicating that these two values are equal and hence the game has a stable solution.

Part (c) Most candidates realised that R_1 dominated R_3 and hence Rob should never play R_3 .

Part (d) To find the optimal mixed strategy, when a letter such as p is introduced, there should be an indication that this is the probability that Rob is choosing R_1 , for example. Three expressions in p were often written down with no indication as to what they represented. Although three linear graphs were often drawn, many candidates chose the wrong pair to solve for the optimal strategy and seemed to be guessing rather than reasoning correctly which point to select. Those who obtained the correct optimal strategy were usually able to find the value of the game.

Q14.

Despite the standard nature of part (a)(i), it was surprising to see many candidates unaware of how to find the optimal mixed strategy. A good sketch showing the feasible region was expected with the highest point of the feasible region being selected in order to find the probability of playing the various rows. Many candidates gave no reason for introducing a probability p nor explained what the actual mixed strategy was when the value of p had been found.

In part (a)(ii), it was quite surprising to see a number of candidates substituting $p = 0.5$ into their third (unused) expression and consequently obtaining the wrong value of the game, even though the answer was printed in the question.

In part (b)(i), almost everyone obtained the correct answer of $1 - p - q$. Part (b)(ii) was more discriminating, and it was obvious that some were totally unfamiliar with the technique needed. Many did not see the need to use the value of the game, and others made errors in their algebra. Nevertheless, there were quite a few completely correct solutions to this part of the question.

Q15.

In part (a), in order to show that the game has a stable solution, it was expected that the minimum values in the rows and maximum values in the columns would be indicated before finding the maximum of the minima and the minimum of the maxima. Some statement should then have been made indicating that these two values are equal and hence the game has a stable solution. To find the optimal mixed strategy in part (b), when a letter such as p is introduced, there should be an indication that this is the probability

that Ros is choosing R_1 , for example. Three expressions in p were often written down with no indication as to what they represented. Although three linear graphs were often drawn, many candidates chose the wrong pair to solve for the optimal strategy and seemed to be guessing rather than reasoning correctly which point to select. Those who obtained the correct optimal strategy were usually able to find the value of the game.

Q16.

Part (a)(i) proved to be the most testing part of this question, with very few candidates giving a good reason why Rose should choose R_1 as the play-safe strategy. The minimum value in each row had to be calculated and then the maximum of these minima needed to be shown to equal -1 . In part (a)(ii), the better solutions gave a table with the column maxima 5, 2 and 5 clearly stated and with the minimum of these values indicated. A statement that the minimax value, 2, was not equal to the maximin value, -1 , from part (a)(i) was then required to show that the game has no stable solution.

In part (b), most candidates realised that R_1 dominates R_3 and showed that corresponding entries of R_1 were greater than those of R_3 .

The expressions for the expected gains in part (c) were usually correct, although some made algebraic slips when trying to simplify their expressions. There was a casual approach by many when drawing the graphs. Lines were often too short or too long and many made no attempt to indicate values on the vertical axis. Candidates do not appear to know what the three lines on the graph represent. Some seemed to guess which two expressions to equate when finding the optimal mixed strategy for Rose, instead of considering the highest point in the feasible region.

It was necessary to make a statement indicating that Rose plays R_1 for $2/3$ of the time and R_2 for $1/3$ of the time in order to score full marks in part (c)(iii). It was quite surprising to see a number of candidates substituting $p = 2/3$ into their third (unused) expression and consequently obtaining the wrong value of the game.

Q17.

The idea of dominance seems to be well understood. Marks were lost in part (b) for not identifying the actual play safe strategies. It is not sufficient to draw circles around a couple of numbers and to simply state there is no saddle point. Unfortunately, this is precisely what many candidates offered as their solution. This question was a simple example of game theory, reducing to a 2 by 2 game, and most candidates wrote down and solved correctly an equation in p and hence the expected points gain for Sam. Consequently, it was pleasing to see many high marks in this question.

Q18.

The idea of dominance seems to be well understood but only the best candidates could explain the term 'zero-sum game'. In order to show that the game has no stable solution, it is expected that the minimum values in the rows and maximum values in the columns be indicated before finding the maximum of the minima and the minimum of the maxima. Some statement should then be made indicating that these two values are not equal and hence the game has no stable solution. To find the optimal mixed strategy, when a letter such as p is introduced, there should be an indication that this is the probability that Rowan is choosing R_2 for example. Three expressions in p were often written down with no indication as to what they represented. When three linear graphs had been drawn, many candidates made mistakes in choosing the appropriate highest point of the feasible region. Again, this gave the impression that many were following a recipe rather than understanding what this selection meant. Those who obtained the correct optimal strategy

were able to find the value of the game.

Q19.

Part (a) It was necessary to work out the row minima and column maxima and then to **show** that the largest entry in the row minima was not equal to the least entry of the column maxima in order to demonstrate that the game had no stable solution. Candidates need to realise that all these steps in the working need to be shown in order to earn full marks.

Part (b) It was expected that candidates would have shown that elements of P_1 dominated corresponding elements of P_3 and so Pat should never play strategy P_3 .

Part (c) Most candidates attempted to find expected gains corresponding to Quigley playing the different strategies. These were usually calculated with Pat choosing P_1 with probability p and P_2 with probability $1 - p$, although some worked effectively by reversing these probabilities. It is advisable to state what p represents if this variable is introduced. Many of the graphs of expected gains were not drawn accurately enough so as to identify the critical point of intersection. Very few graphs had axes labelled showing the expected gain or payoff for Pat on the vertical axis and p ranging from 0 to 1 on the horizontal axis. It would seem that several candidates were selecting two of the three lines quite arbitrarily in order to find the value of p and hence the optimal mixed strategy for Pat.

Some candidates had been taught a method which produced the mirror image of the expected graph, but there was often a reluctance on the part of these to indicate whether p ranged from 1 to 0 or Pat was choosing P_1 with probability $1 - p$. Consequently, it was difficult to know whether these candidates really understood what they were doing or whether they were simply drawing graphs using the numbers from the table.

Q20.

Parts (a) and (b) of this question were well answered. However a small number of weaker candidates used $p - 1$, rather than $1 - p$ as one of their probabilities. In part (c) many candidates correctly used p , q and $(1 - p - q)$ for the three probabilities but still found the completion of the question difficult. There were however a number of outstanding solutions to the question. Candidates who used dominance to reduce the problem to one involving a 2×2 matrix were in general successful.

Q21.

Part (a) was well answered and part (b) was a good discriminator. Candidates who used the correct approach of letting the play for the two strategies to be p and $(1 - p)$ found the completion of the question straightforward. There were however a number of candidates who used p and $(p - 1)$. Some candidates failed to state the value of the game which was a waste of a mark.

Q22.

Again this question was well answered, with many complete solutions to a conceptually difficult question. This was the first time that a game theory question had been set in which the values were algebraic. Some candidates solved for p first, others for x , with equal success.

Q23.

Candidates knew the theory for part (a) of this question but there were very few scripts that earned full marks. Candidates had to show calculations to answer this question rather than simply stating that there is no play safe solution.

Part (b) was well answered.

Part (c) was the least well answered question on the paper. Some candidates adopted the correct approach, using probabilities p , q , and $1 - p - q$, but still found the completion of the question difficult. There were, however, a number of correct solutions.

Q24.

There were some excellent answers to this question, with candidates showing a clear understanding of game theory. However, some candidates were less well prepared. Despite drawing correct diagrams, they were unable to identify the correct solution to the problem, and a value of 0.5 for p was frequently obtained.

Q25.

Parts (a) and (b) were both well answered. However, a number of candidates managed to reduce the 2×3 matrix into a 2×2 matrix. Part (c) was well answered by the majority of the candidates. There were two approaches to solving the equations, algebraically or graphically, with both methods proving to be successful. A few candidates initially found the optimal mixed strategy for B , found the value of the game and then found the mixed strategy for A , but they still managed to finish the paper in the allotted time.

Q26.

Although this was a challenging question it was pleasing to see many excellent solutions. To find the value of p in part (b) some candidates solved the equations algebraically and others graphically, with both methods proving to be successful. A few candidates thought that they only had to find the strategy for player A .

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